Introduction

This activity involved the design, construction, and testing of an aircraft wing box structure. An aircraft wing was found to be experiencing compressive, tensile, and shearing forces during working conditions. The wing box is the primary support structure that forms a wing. It is made up of the upper and lower skin, stringers, spars, ribs, and butt straps, constructed using greyboard and rivets in this project. Design parameters, such as thickness and rib spacing, were chosen to meet requirements whilst reducing cost. After completion, the wing box was tested under a four-point bending arrangement. Performance of the wing box structure can be compared to the reserve factors to see if the structure performed as expected.

Load - extension plot

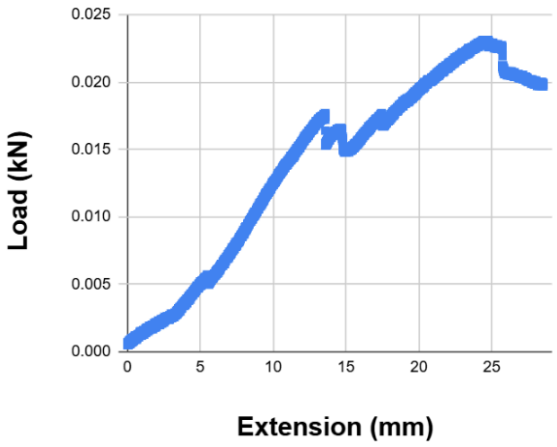


Figure 1 - The load extension plot of the wing box

Figure 1 shows that the ultimate failure mode of the wing box occurred at ~0.023 kN under an extension of ~24 mm. Other non-critical failures occur at lower extensions, such as at ~ 0.005 kN, which was likely due to a stringer breaking, and at ~0.017 kN.

Comparison With Calculated RF Values

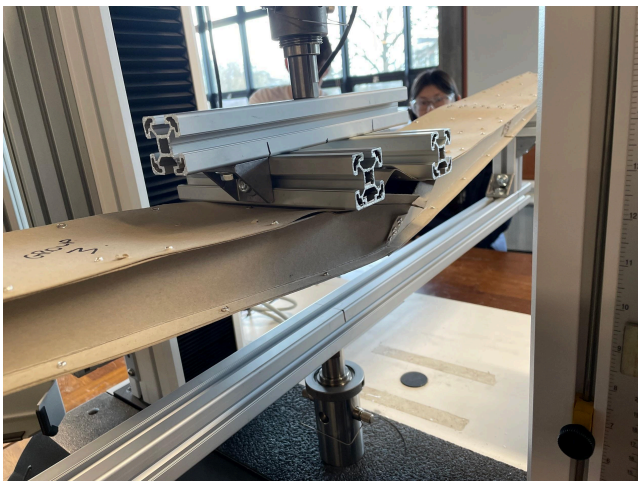


Figure 2 - The buckling of the wing box at ultimate failure load

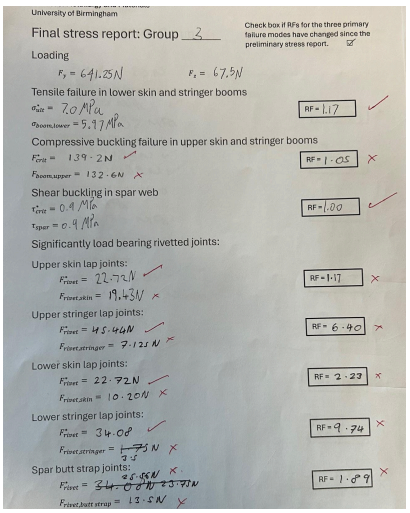


Figure 3 - Final stress report

From Figure 2 we can determine that the wing box failed by spar webbing and upper string buckling. Comparing the calculated RF values in Figure 3, this is what should be expected due to these failure modes having the lowest RF values.

Improvements to the model and method

Team	Ultimate Load (N)	Mass (g)	Material mass purchased (g)	"Buy to Fly" ratio	Cost (£)	Performance Figure of Merit
A1	18.9	487	1208	2.5	127.92	16.9
A2	8.7	440	1556	3.5	179.86	11.0
A3	23.1	509	1586	3.1	185.12	11.1
A4	8.7	510	1919	3.8	192.82	4.0
A5	12.2	485	1688	3.5	203.02	6.7
A6	1.8	545	2160	4	257.2	0.5

Figure 4 - Summary of the performance metrics

One immediate improvement that would increase the ultimate failure load is if we implemented the ribs that were created. This would reduce the area over which a buckling force can act and transfer the point loads from the testing system.

A change to the method that could be used is being more space efficient with the cardboard that is used, due to only 509g of the 1586 g purchased material being used from figure 4, leading to an efficiency of 32%, which is imperfect. This could be reduced by adding the ribs, which contributed massively to the wasted material

Another area of improvement was the order of assembly. While assembling the wingbox, we first attached the spars to the upper skin. This made it crowded and difficult to attach the stringers without the rivets falling out, due to the curvature of the upper skin. If the project were to be completed again, the assembly would be improved if we first attached the ribs to the spars instead of the upper skin. If we then attached the skins with the stringers around the spars, then the crowded nature of the structure would be reduced

Skills Reflection

Teamwork and cohesion were developed during this project as sub-teams worked individually to produce components as a group to then assemble these parts, working with other sub-groups. Teamwork was also important in the design and calculation stage, sharing results and ideas, to ensure the final design worked with the chosen numbers of ribs and stringers, thicknesses, and rib spacing. This also required communication, sharing the important information clearly to make the process efficient and accurate, so that all sub-groups had the necessary values and a set plan for the team.

To decide the ideal parameters, it required unfamiliar calculations, which has developed our complex problem-solving and resilience. It has evidently improved resilience as there were many setbacks during this process, but as a team we found and chose our design, again highlighting teamwork. This complex exercise has provided knowledge of how to approach such situations in the future, trying multiple ideas to reach the best solution.

This project required time management to produce the wingbox on time for testing, to deal with this the project was split into a design session and two build sessions. We set the first build session to build the individual components and the second was to assemble the wingbox. This shows a good time management plan, however, this didn't go as planned due to some clashes between sub-groups and unexpected workloads. This led to the final product not having all the ribs in place, highlighting time management as a skill that can be improved further in the group.